Θέματα Στατιστικής στη γλώσσα R

- Η εμπειρική αθροιστική συνάρτηση κατανομής
- Γραφικός έλεγχος καλής προσαρμογής – qqplot
- Διαστήματα εμπιστοσύνης
  - ✓ η ερμηνεία του σ.ε.
  - ✓ η σχέση με το μέγεθος του δείγματος
  - ✓ για τη μέση τιμή κανονικής
    - ❖ διασπορά γνωστή
    - ❖ διασπορά άγνωστη

Η εμπειρική α. σ. κ.

Έστω $\underset{\sim}{X} = (X_1, \dots, X_n)$ τ.δ. από κατανομή με α.σ.κ. $F: \mathbb{R} \to [0,1]$

Η συνάρτηση $F_n(x) = \frac{\sum_{i=1}^{n} \mathbb{1}_{(-\infty, x)}(X_i)}{n} \equiv \frac{\#\text{ των } X_i \le x}{n}$, $x \in \mathbb{R}$

ονομάζεται εμπειρική αθροιστική συνάρτηση κατανομής.

Είναι και αυτή τυχαία μεταβλητή και ισχύει ότι:

- $\mathbb{1}_{(-\infty, x)}(X_i) \sim \text{Bernoulli}(F(x))$ $i = 1, \dots, n$ και είναι ανεξάρτητες

  όπου $\mathbb{1}_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$ δείκτρια συνάρτηση

- $W_x^n = \sum_{i=1}^{n} \mathbb{1}_{(-\infty, x)}(X_i) \sim \text{Binomial}(n, F(x))$

- $n F_n(x) \sim \text{Binomial}(n, F(x))$

Η εμπειρική α. σ. κ.

Οπότε για την εμπειρική α.σ.κ. $F_n(x)$ της $F(x)$ ισχύουν:

- $E(F_n(x)) = F(x) \quad \forall x \in \mathbb{R}$
- $\text{var}(F_n(x)) = \frac{F(x)(1-F(x))}{n} \quad \forall x \in \mathbb{R}$

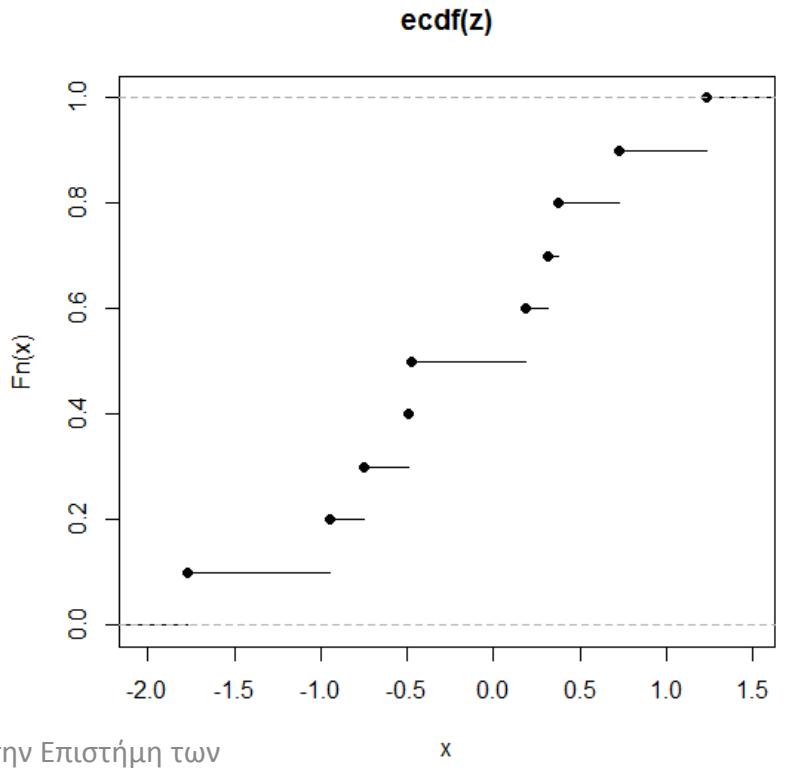
Δηλαδή η $F_n(x)$ είναι αμερόληπτη εκτιμήτρια της $F(x)$ και επειδή $\lim_{n\to\infty} \text{var}(F_n(x)) = 0$ είναι επιπλέον "συνεπής"

$$F_n(x) \xrightarrow[n\to\infty]{\text{κ.π.}} F(x) \quad \forall x \in \mathbb{R}.$$

έχω δηλαδή σύγκλιση κατά πιθανότητα.

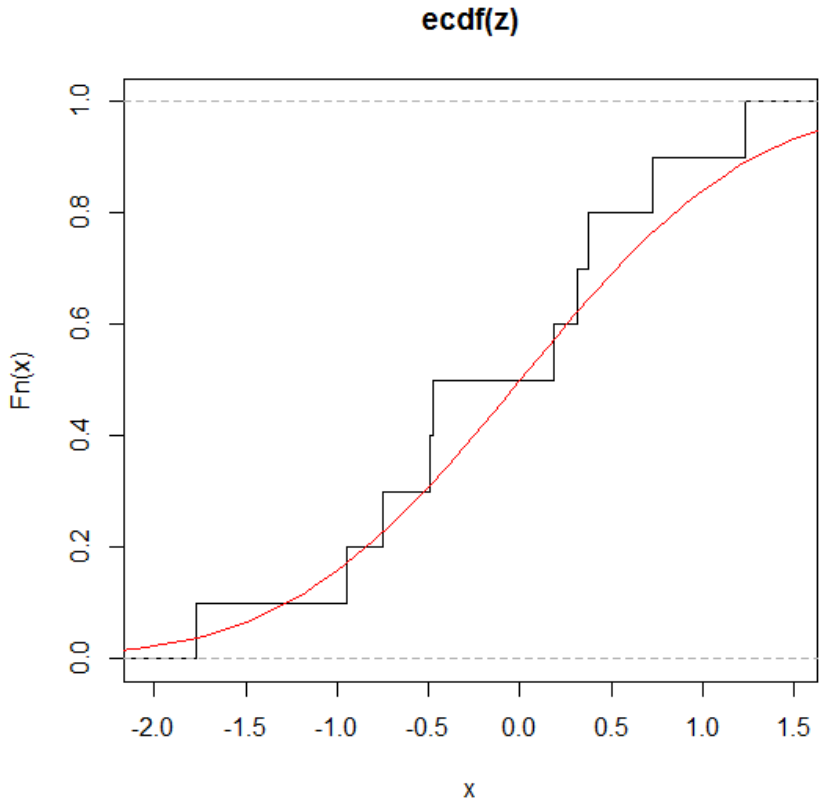
Κώδικας-ecdf

```
> z<-rnorm(10)
> ecdf(z)
Empirical CDF
Call: ecdf(z)
 x[1:10] = -1.7681, -0.94457, -0.74734,  ..., 0.72241, 1.2328
> summary(ecdf(z))
Empirical CDF:     10 unique values with summary
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-1.7680 -0.6836 -0.1458 -0.1604  0.3567  1.2330
> z[1]
[1] 0.3147419
> ecdf(z)(z[1])
[1] 0.7
> which(sort(z)==z[1])
[1] 7
> sort(z)[1]
[1] -1.768119
> ecdf(z)(sort(z)[1])
[1] 0.1
> ecdf(z)(-2)
[1] 0
> ecdf(z)(3)
[1] 1
> plot(ecdf(z))
```



Κώδικας-ecdf

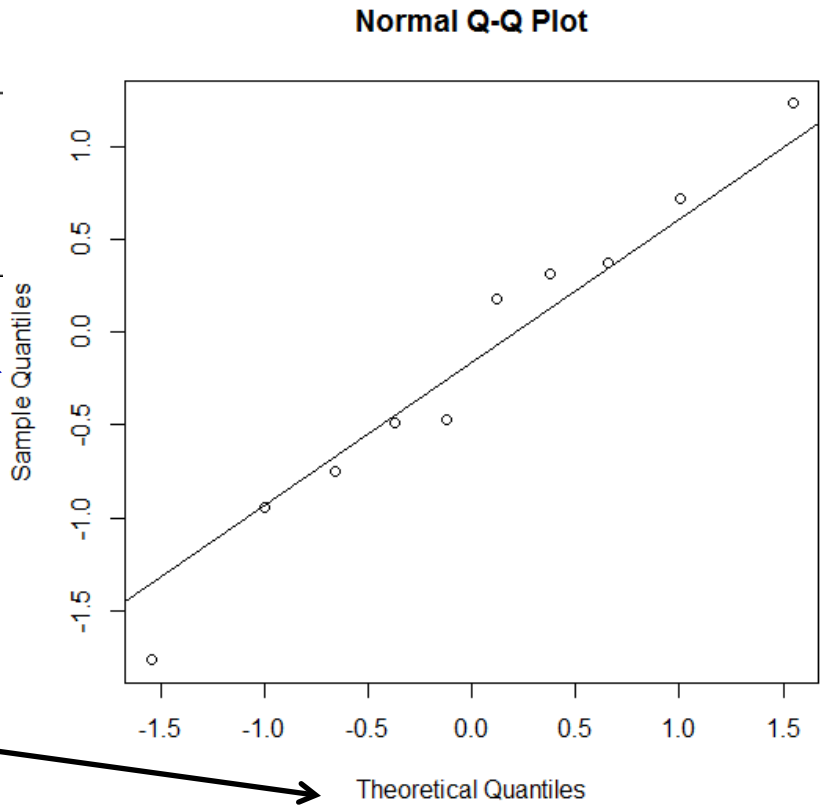
```
> Fx<-ecdf(z)
> plot(Fx, do.points=FALSE, verticals=TRUE)
Waiting to confirm page change...
> plot(pnorm, add=TRUE, xlim=c(-5,5),col="red")
```



Οπτικός έλεγχος κανονικότητας-Κώδικας

```
> qqnorm(z)
Waiting to confirm page change...
> qqline(z)
```

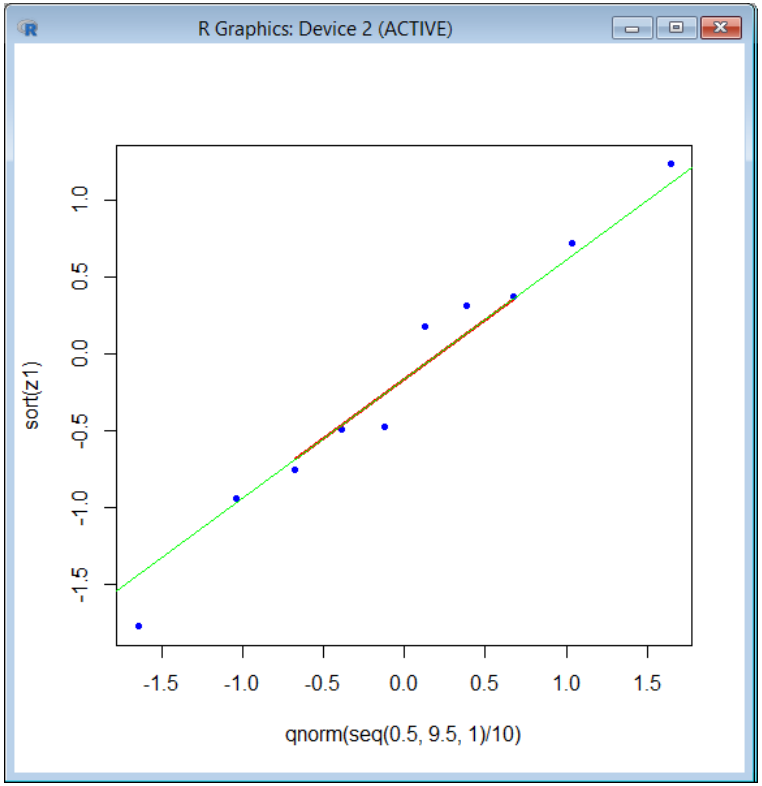
Ordered observations $x_{(j)}$	Probability levels $(j - \frac{1}{2})/n$	Standard normal quantiles $q_{(j)}$
[1] -1.77	.05	−1.645
[2] -0.94	.15	−1.036
[3] -0.75	.25	−.674
[4] -0.49	.35	−.385
[5] -0.47	.45	−.125
[6] 0.18	.55	.125
[7] 0.31	.65	.385
[8] 0.37	.75	.674
[9] 0.72	.85	1.036
[10] 1.23	.95	1.645




```
> round(qnorm(seq(0.5,9.5,1)/10),3)
 [1] -1.645 -1.036 -0.674 -0.385 -0.126  0.126  0.385  0.674  1.036  1.645
```

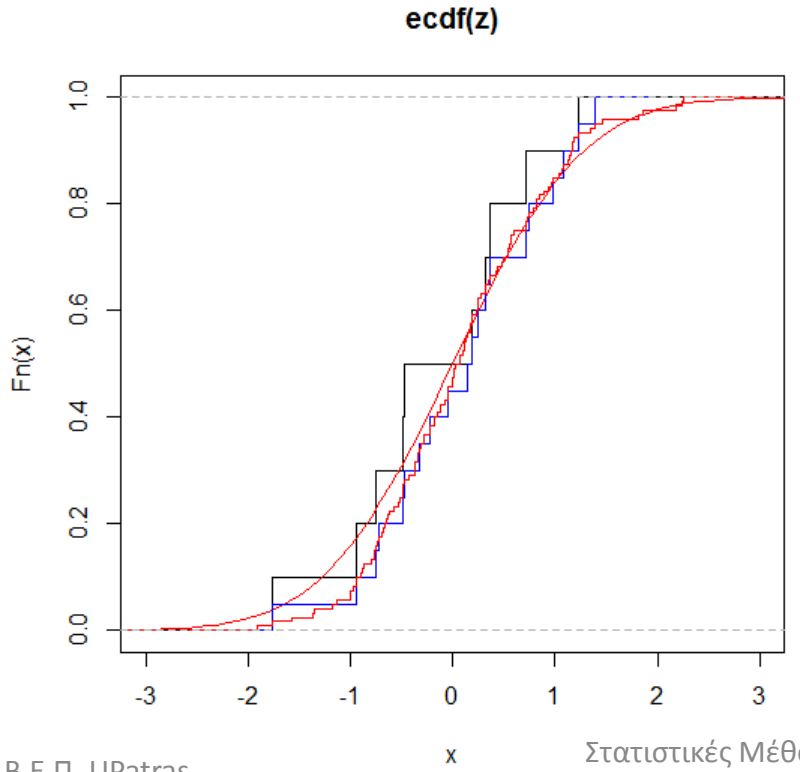
Οπτικός έλεγχος κανονικότητας-Κώδικας

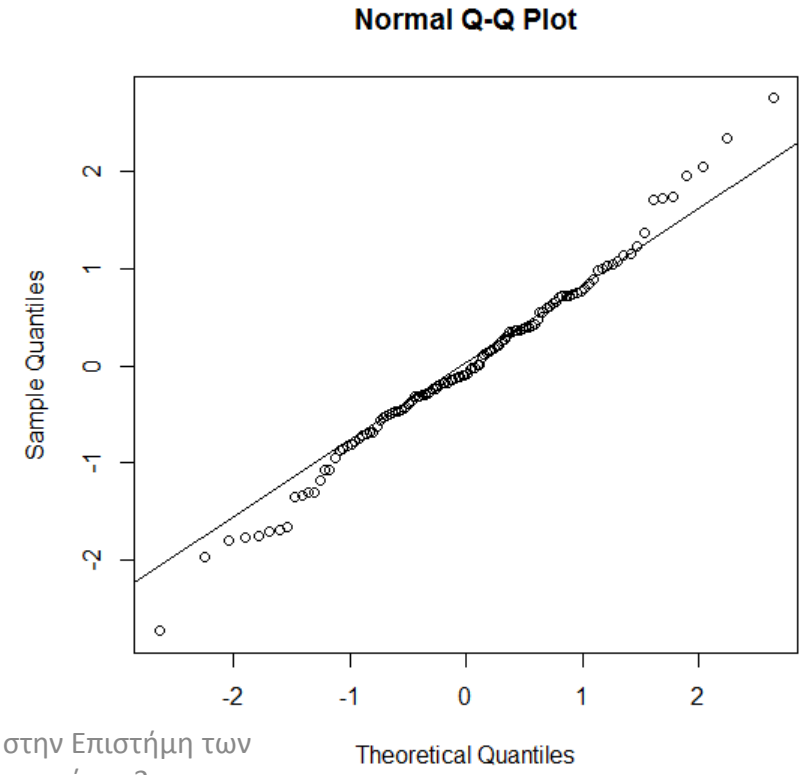
```
> plot(qnorm(seq(0.5,9.5,1)/10),sort(z1),pch=20,col="blue")
> lines(qnorm(c(0.25,0.75)),quantile(z1,probs=c(25,75)/100),
+       type="l",lwd=2,col="red")
> slope<-(quantile(z1,probs=0.75)-quantile(z1,probs=0.25))/
+ (qnorm(0.75)-qnorm(0.25))
> intercept<-quantile(z1,probs=0.75)-slope*qnorm(0.75)
> abline(a=intercept,b=slope,lwd=1.5,col="green")
```



Κώδικας

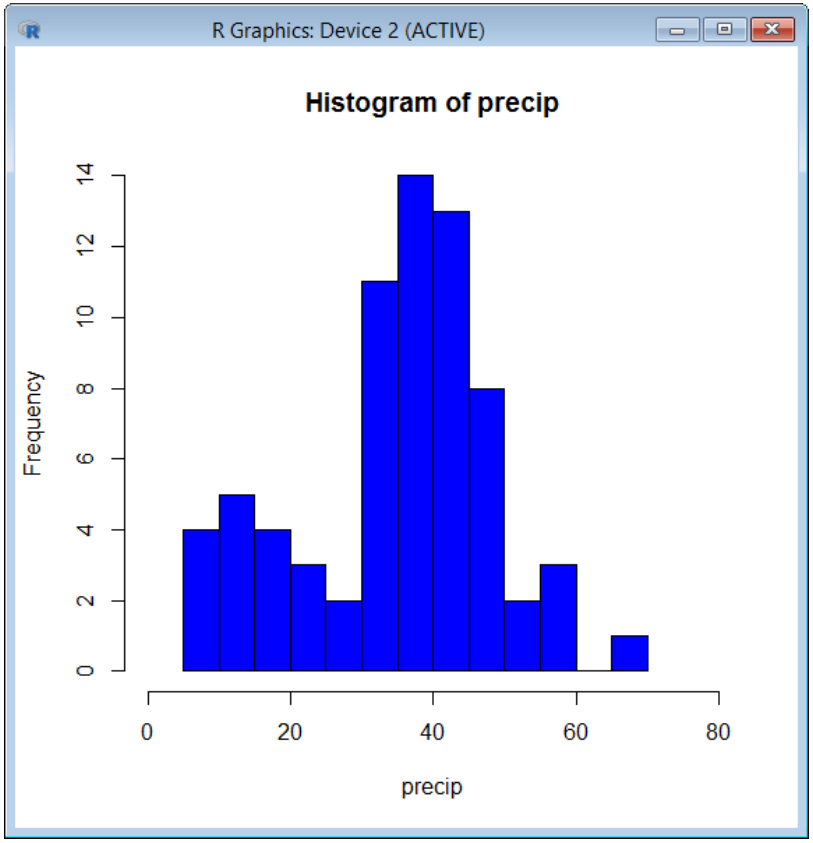
```
> z<-c(z1,rnorm(10))
> plot(ecdf(z), do.points=FALSE, verticals=TRUE,add=TRUE,col="blue")
> z
  [1]  0.3147418597 -0.4726607352  0.3706207496 -0.7473350460 -0.4925296902
  [6]  0.7224123415  1.2327545049 -0.9445736881 -1.7681187035  0.1810586228
 [11] -0.3183389678  0.1504133181  0.7414939817  1.3947300882 -0.2243841737
 [16] -0.7203366137 -0.0529727192  1.0752820107  0.2452762116  0.9826648323
> z<-c(z,rnorm(100))
> plot(ecdf(z), do.points=FALSE, verticals=TRUE,add=TRUE,col="red")
```

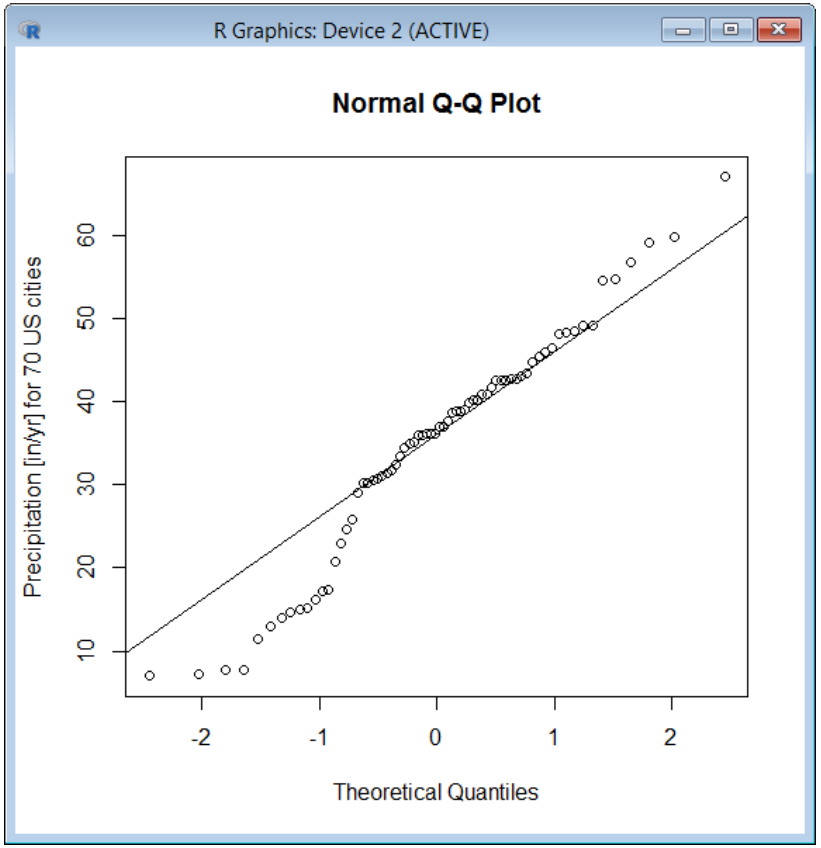




Κώδικας-Παράδειγμα (δεδομένα ύψους βροχής)

```
> hist(precip,nclass=20,xlim=c(0,80),col="blue")
> qqnorm(precip, ylab = "Precipitation [in/yr] for 70 US cities")
> qqline(precip)
```

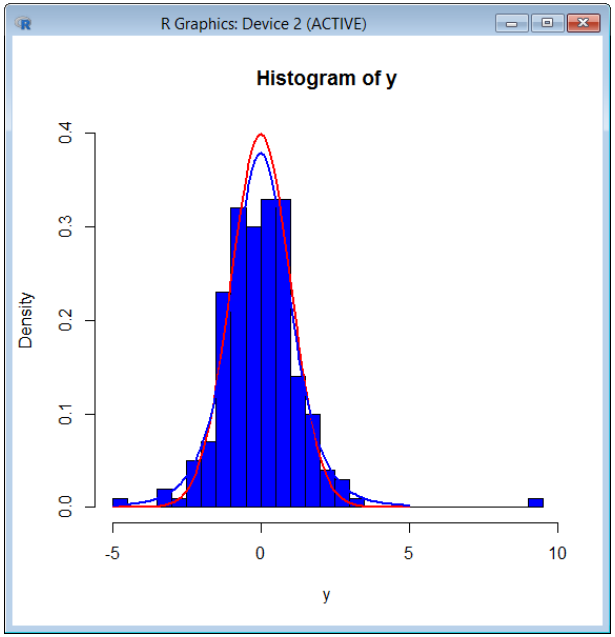


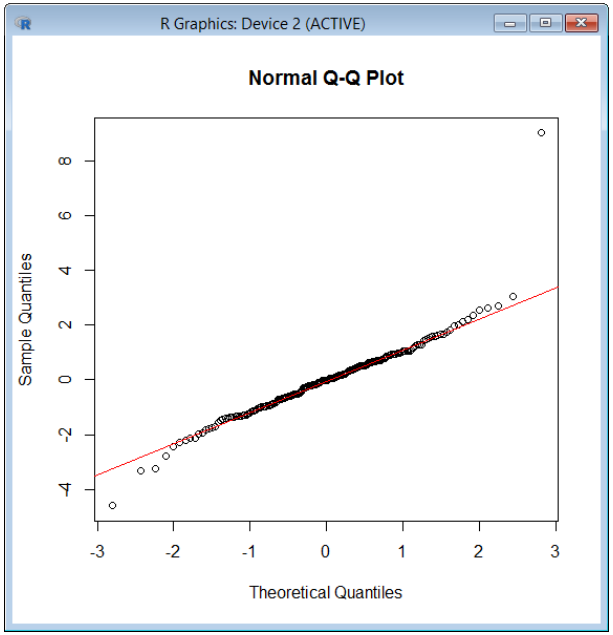


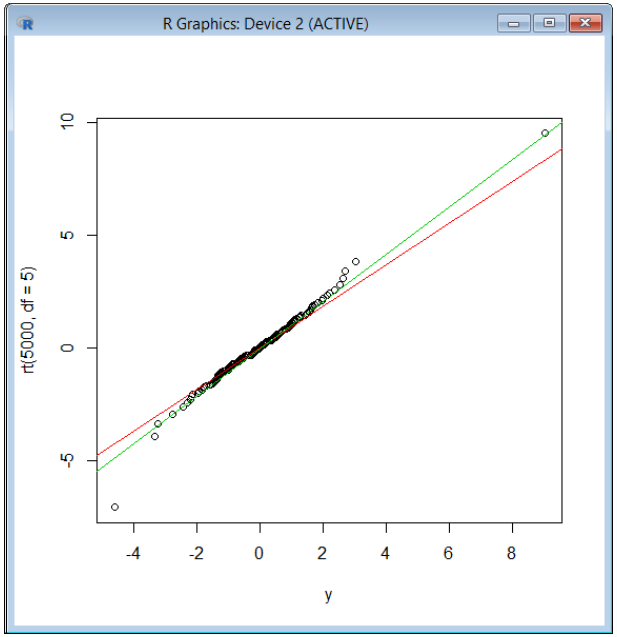
Οπτικός έλεγχος καλής προσαρμογής

σε άλλες κατανομές (Student's t)-Κώδικας

```
> y <- rt(200, df = 5)
> qqplot(y, rt(300, df = 5))
Waiting to confirm page change...
> qqline(y, distribution = function(p) qt(p, df = 5),
+        prob = c(0.1, 0.6), col = 2)
> qqline(y, distribution = function(p) qt(p, df = 5),
+        prob = c(0.25, 0.75), col = 3)
```



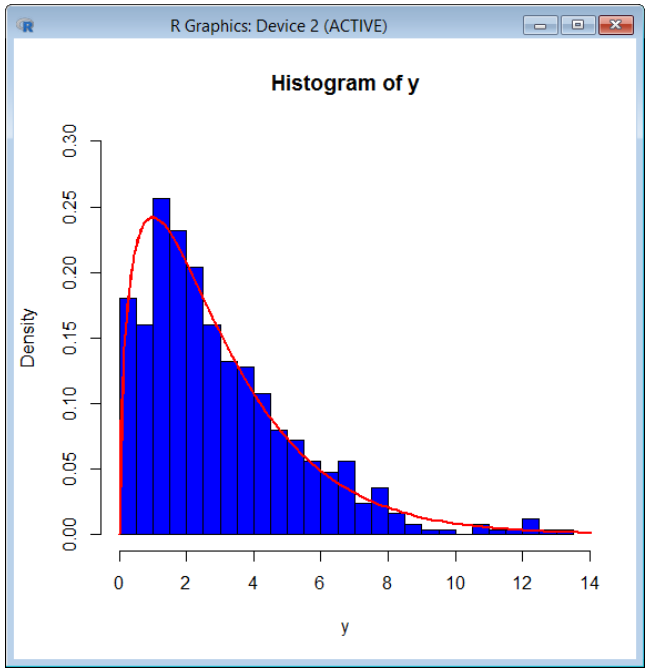


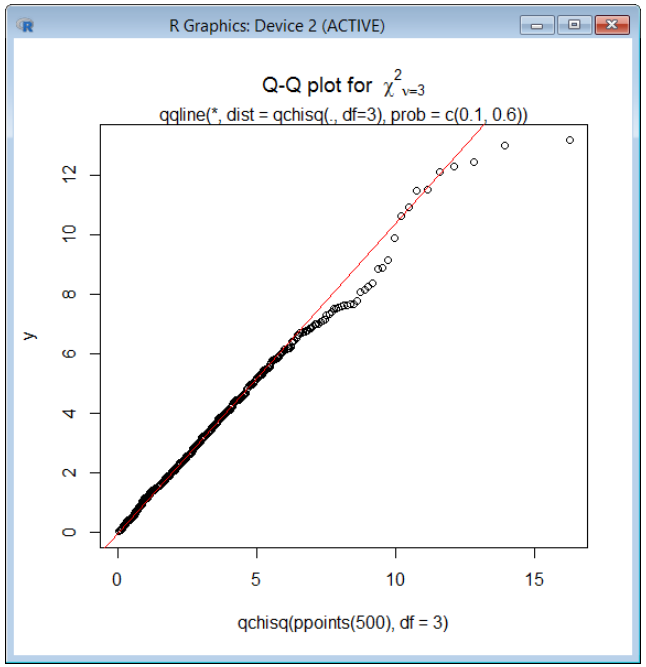


Οπτικός έλεγχος καλής προσαρμογής

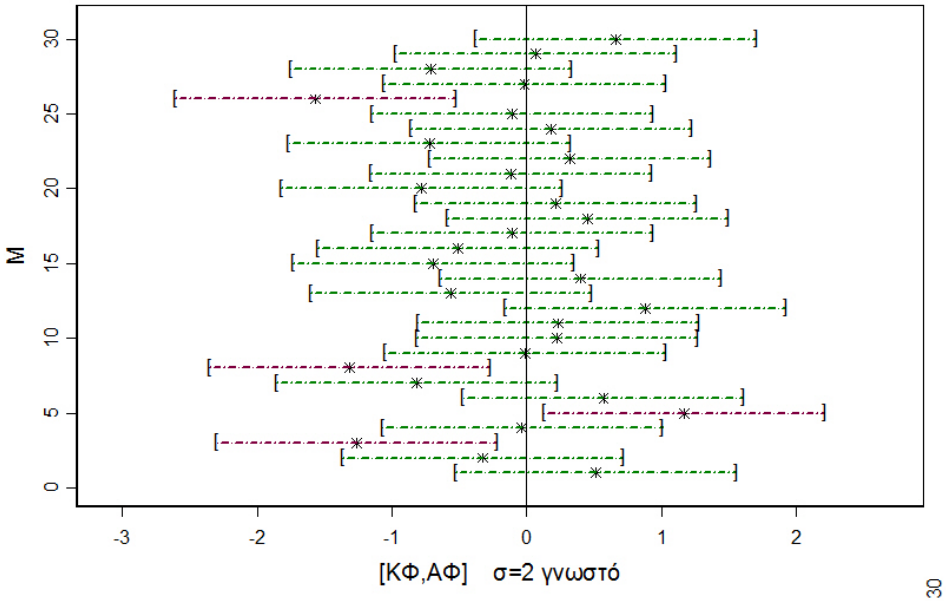
σε άλλες κατανομές (X^2)-Κώδικας

```
> y <- rchisq(500, df = 3)
> ## Q-Q plot for Chi^2 data against true theoretical distribution:
> qqplot(qchisq(ppoints(500), df = 3), y,
+        main = expression("Q-Q plot for" ~~ {chi^2}[nu == 3]))
Waiting to confirm page change...
> qqline(y, distribution = function(p) qchisq(p, df = 3),
+        prob = c(0.1, 0.6), col = 2)
> mtext("qqline(*, dist = qchisq(., df=3), prob = c(0.1, 0.6))")
```

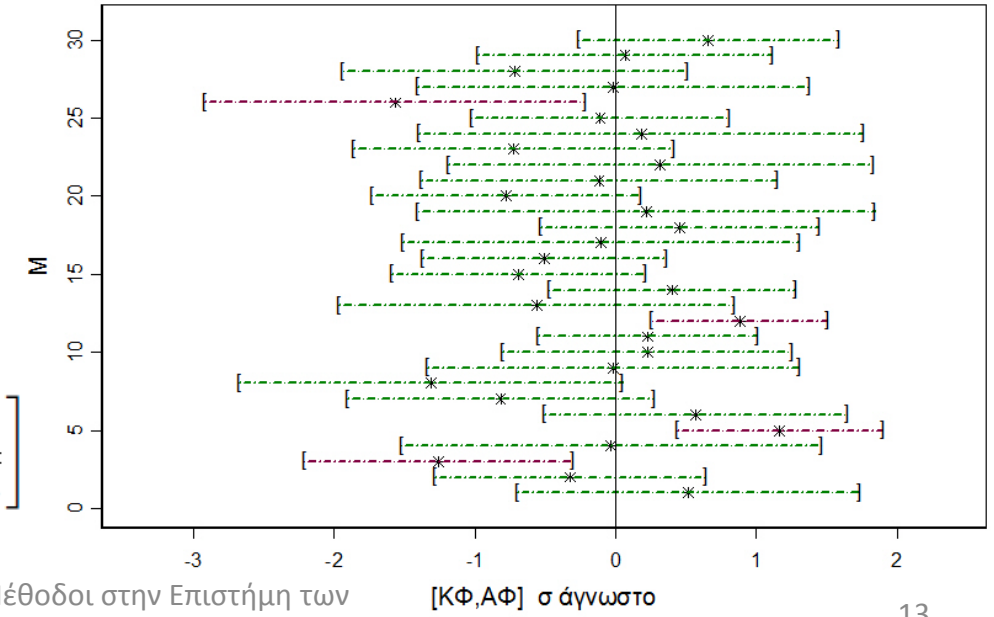




Διαστήματα Εμπιστοσύνης



$$\left[\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$



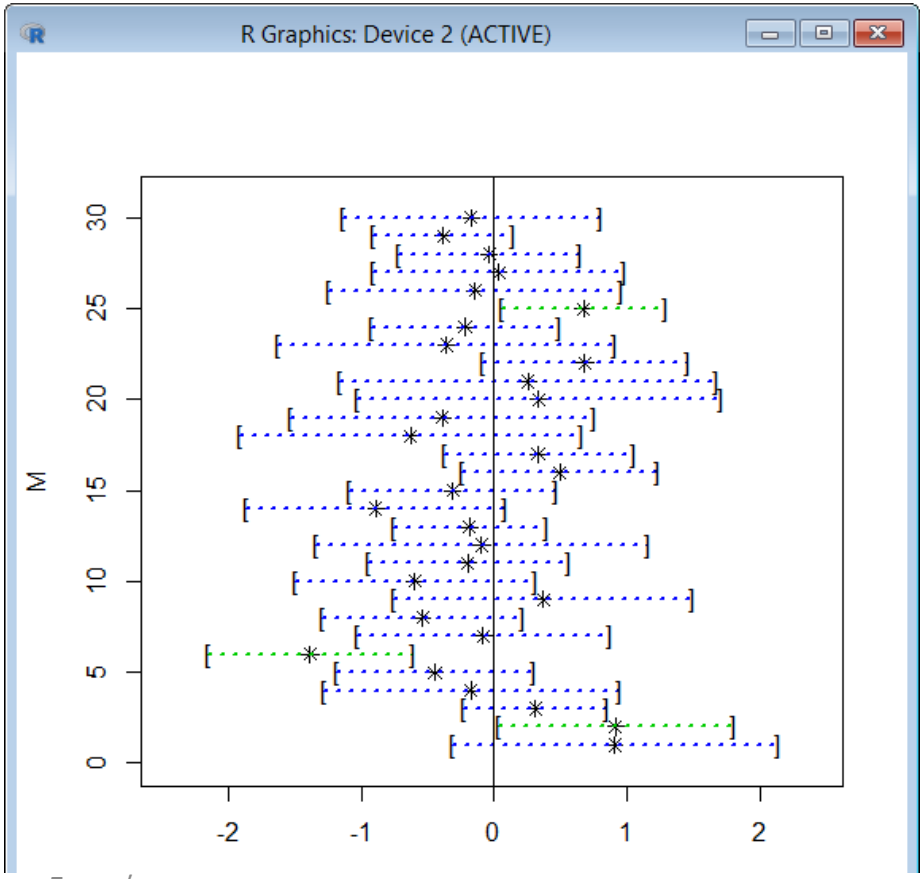
$$\left[\overline{X} - t_{n-1}(\alpha/2)\frac{S}{\sqrt{n}},\ \overline{X} + t_{n-1}(\alpha/2)\frac{S}{\sqrt{n}}\right]$$

Διαστήματα Εμπιστοσύνης: η ερμηνεία του σ.ε.

```
> s
 [1] 1.819709 1.654812 1.352845 1.052756 1.397919 1.785158
 [7] 1.794624 1.260199 1.748311 1.126880 1.678715 1.615690
[13] 1.466358 1.387275 1.885143 1.164992 1.100625 1.750822
[19] 1.776309 1.914089 1.936415 1.056705 1.807750 1.276713
[25] 1.689612 1.441474 1.533761 1.631297 1.049493 1.491035
```

```
> N2(30,10,0.1,0,s)
Waiting to confirm page change..
 [1] 2.441024 1.771613 1.081134
 [4] 2.218797 1.488548 1.542167
 [7] 1.895853 1.503706 2.238480
[10] 1.805696 1.507070 2.483619
[13] 1.138120 1.940759 1.549560
[16] 1.462575 1.419440 2.562162
[19] 2.284833 2.727073 2.821752
[22] 1.526949 2.532931 1.409244
[25] 1.236595 2.193413 1.878055
[28] 1.360058 1.051815 1.926961
```

εύρος των διαστημάτων που κατασκευάστηκαν

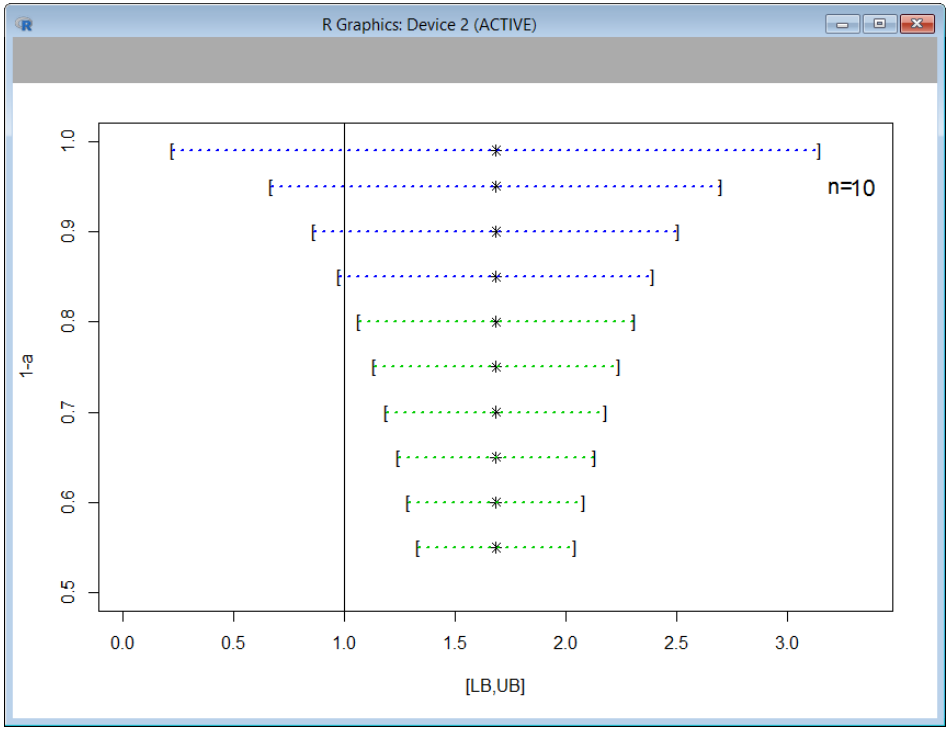


Κώδικας

```
> N2<-function(M,n,a,m,s) {
+ y<-array(rep(0,2*M),dim=c(M,2))
+ for (i in 1:M) {
+ x<-rnorm(n,m,s[i])
+ y[i,1]<-mean(x)+qt(a/2,n-1)*sqrt(var(x)/n)
+ y[i,2]<-mean(x)+qt(1-a/2,n-1)*sqrt(var(x)/n)
+ }
+ for (i in 1:M) {
+ if  (i==1) {
+ plot((y[1,1]+y[1,2])/2,1,pch=8,xlim=c(min(y[,1])-0.3,max(y[,2])+0.3)
+ }
+ if ((y[i,1]>m)||(y[i,2]<m)){
+ points((y[i,1]+y[i,2])/2,i,pch=8,col=1)
+ lines(c(y[i,1],y[i,2]),c(i,i),lty = 3,lwd=2.2,col=3)
+ text(y[i,1],i,"[",col=1)
+ text(y[i,2],i,"]",col=1)}
+ if ((y[i,1]<m)&(y[i,2]>m)){
+ points((y[i,1]+y[i,2])/2,i,pch=8,col=1)
+ lines(c(y[i,1],y[i,2]),c(i,i),lty = 3,lwd=2.2,col=4)
+ text(y[i,1],i,"[",col=1)
+ text(y[i,2],i,"]",col=1)}
+ }
+ abline(v = m,col=1)
+ print(y[,2]-y[,1])
+ }
> s<-runif(30,1,2)
```

Διαστήματα Εμπιστοσύνης: η ερμηνεία του σ.ε.

```
> x
 [1] 5.09787041 2.28965487 2.43218122 0.86540021
 [5] 0.91974851 0.55373450 1.28536466 1.22803143
 [9] 0.08675835 2.09133592
> c(mean(x),var(x))
[1] 1.685008 2.021856
```



Διαστήματα Εμπιστοσύνης και μέγεθος δείγματος

